BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

CPU Performance and Embedded Systems

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. What does the Program Counter hold?

2. A = 1 and B = 0. What is A AND B?
   A) 1
   B) 2
   C) 0
   D) 10

3. "A NIC's purpose is keeping files when power is off." Correct this. Explain why.

4. Marks must be integers from 0 to 100. Classify 50, 101 and text 'fifty'. Why does accepting 50 not prove it is the pupil's real mark?

Today's objectives

Explain the core ideas of cpu performance and embedded systems using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

Clock speed
The number of CPU cycles per second, commonly measured in GHz.

Number of cores
More cores can process more independent tasks concurrently.

Cache size
Larger cache can reduce slower memory accesses for frequently used data.

Embedded system
A dedicated computer designed into another device for a limited/specific purpose.

BLACK PEN: Write one definition in your own words.

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Learn and practise

New Learning - Key ideas

Clock speed
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Number of cores
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Cache size
Larger cache can reduce slower memory accesses for frequently used data.

Embedded system
A dedicated computer designed into another device for a limited/specific purpose.

I Do - Worked model

Compare CPU A: 2 GHz, two cores with CPU B: 3 GHz, four cores, with other factors unspecified.

- B has more cycles per second.
- B also has more cores for potentially parallel work.
- It may perform better in suitable tasks, but other hardware/software factors matter.
- Do not state that it must always run every program faster.

We Do - Guided practice

1. A washing machine has a controller that operates its cycle. Explain why it is an embedded system.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Calculate how many million cycles a 2 MHz clock completes in three seconds.

```
clockMHz = 2
seconds = 3
millions = ...
print(millions)
```

You Do - Independent exam questions

Q2. State two CPU factors affecting performance. [2 marks]

Q3. Explain why a larger cache may improve performance. [3 marks]

Q4. Give an example of an embedded system and explain why it qualifies. [3 marks]

Exit Ticket - True or False?

More CPU cores can improve performance for tasks that run in parallel.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. The address of the next instruction to be fetched.

2. C - 0

3. The interface that enables a device to connect to a network.

4. 50 is valid. 101 is an integer but outside the range. Text 'fifty' is the wrong type (erroneous). Validation checks rules; it cannot prove the pupil earned 50.

I Do - Explain the model

B has more cycles per second.

B also has more cores for potentially parallel work.

It may perform better in suitable tasks, but other hardware/software factors matter.

Do not state that it must always run every program faster.

We Do - Model answers

1. It performs a specific function rather than general-purpose computing.

Code First - Answer

```
clockMHz = 2
seconds = 3
millions = clockMHz * seconds
print(millions)
```

Expected output: 6

You Do - Self-mark

Q2: Clock speed, cache size, number of cores (any two).

Q3: Frequently needed instructions/data can be accessed faster, reducing waiting for RAM.

Q4: For example a washing-machine controller: built into the device for a dedicated control task.

GREEN PEN: Choose the hardest answer. Correct it on lined paper.